

Task 5. Dominosa

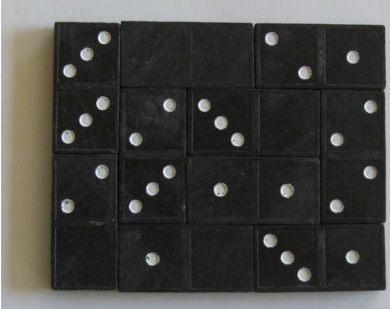
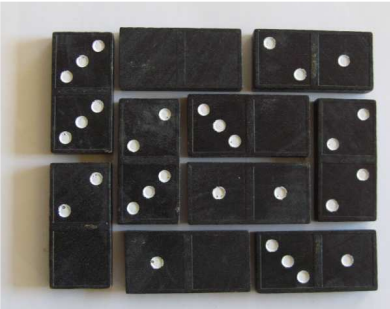
Available marks: 28

The puzzle *Dominosa* was invented in 1874 by O S Adler, named by combining "domino" with his initials. It's very simple: given a grid of integers representing domino pips, determine which number pairs are the actual dominos.

The version of the puzzle to be solved uses just the ten dominoes **0:0 0:1 0:2 0:3 1:1 1:2 1:3 2:2 2:3 3:3**, arranged so that they form a grid of numbers in 4 rows and 5 columns. Your task is to read the numbers and decide where the dominoes lie, by displaying a double-spaced grid with a : symbol between the numbers that form each domino and spaces elsewhere.

Only one solution is required, even if more than one exists.

Example

Arrangement (tight)	Arrangement (expanded)	Data file	Solution
		<pre>30021 32302 23112 01031</pre>	<pre>3 0:0 2:1 : 3 2 3:0 2 : : 2 3 1:1 2 : 0 1:0 3:1</pre>

Assumptions

Every input file contains 4 lines, each with 5 digits, no spaces.

You can assume that each digit in the range 0 to 3 occurs exactly 5 times, but the configuration is not necessarily feasible.

Assessment

As well as the above example (task5sample.txt, used for development only), there are 5 assessable test files of (roughly) increasing difficulty, named task5a.txt, task5b.txt, task5c.txt, task5d.txt and task5e.txt. One of these is an impossible configuration, which your program should be able to detect.

Number of Puzzles solved	Marks Awarded
One	5
Two	11
Three	18
Four	25

3 additional marks are awarded if the impossible case is identified, but only if at least one legitimate puzzle is also solved.

When you believe you have completed this task, show the judges the program and its output.